

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: 6000



Annotate, mark-up, and write on this page as needed.

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

181 in	81.2 in	69.2 in	9.7 in	655 ft-lbs	2.37:1	32 in	1070 kW-h	0.877 in	1.99 in
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Do you think you could have found a better solution given more time?

Probably - we could have weighed more variables for each solution

What, in this round, stopped you from finding a better solution?

time

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

102 in	92.3 in	76.1 in	8.6 in	715 ft-lbs	4.79:1	45 in	990 kW-h	0.733 in	0.28 in
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Do you think you could have found a better solution given more time?

NO - easy to reach a common consensus,
only had 2 plots to analyze

What, in this round, stopped you from finding a better solution?

visualization of the vehicle from the
design variable values

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables

17.8 in	81.1 in	79.1 in	6.8 in	409 lb-ft	2.53:1	47 in	1930 kW-h	0.251 in	0.47 in
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Do you think you could have found a better solution given more time?

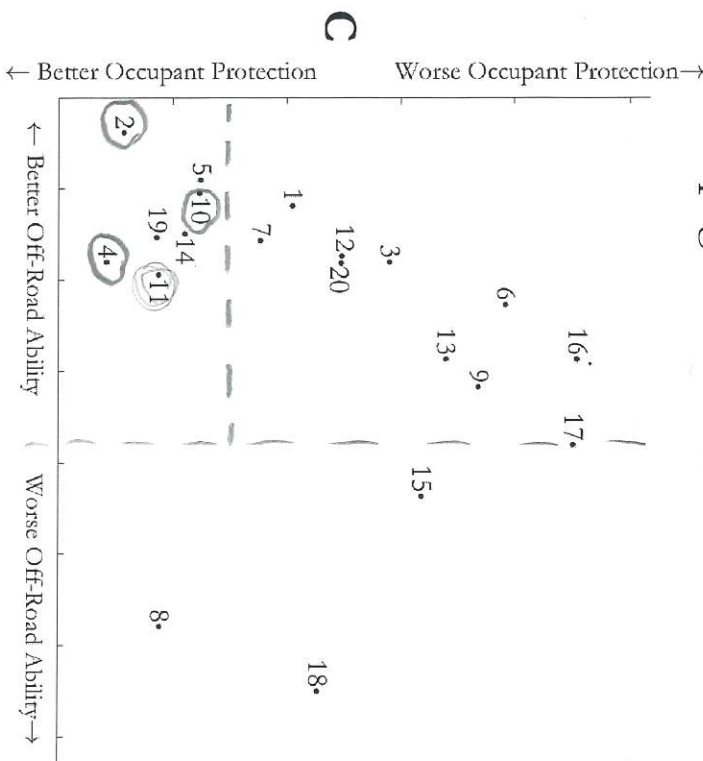
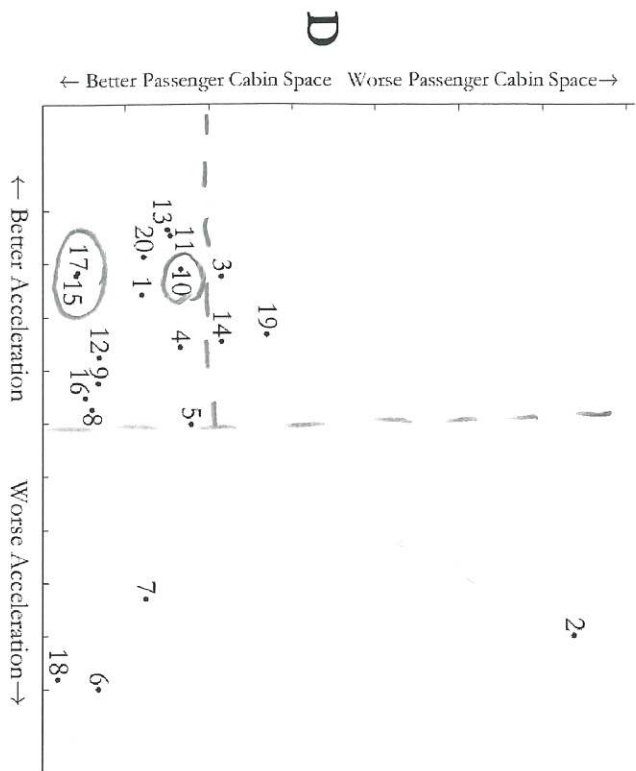
No - we adequately weighed each option
and came to a unanimous decision

What, in this round, stopped you from finding a better solution?

Visualization of the vehicle from the
design variables

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A: 17, 15

A

N0: 2, 7, 6, 18, 15, 8, 17, 5

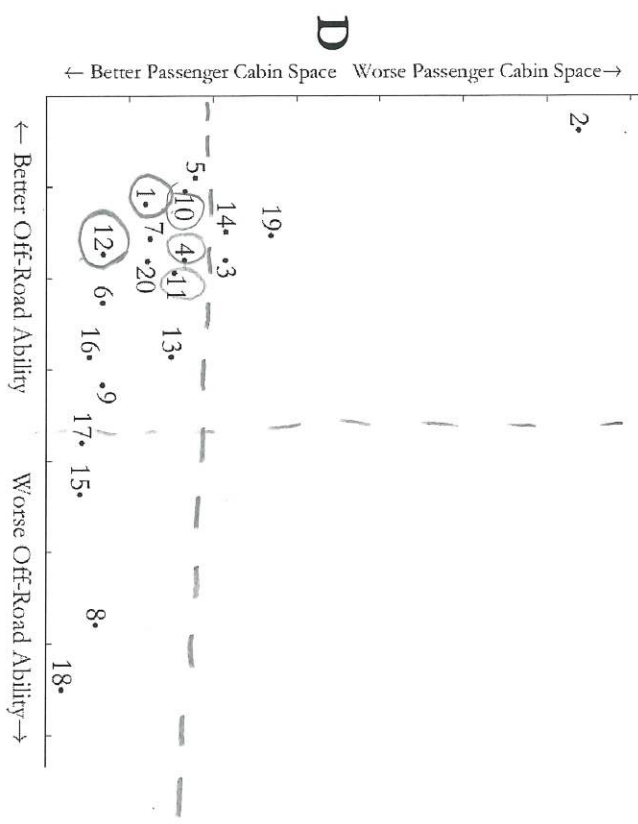
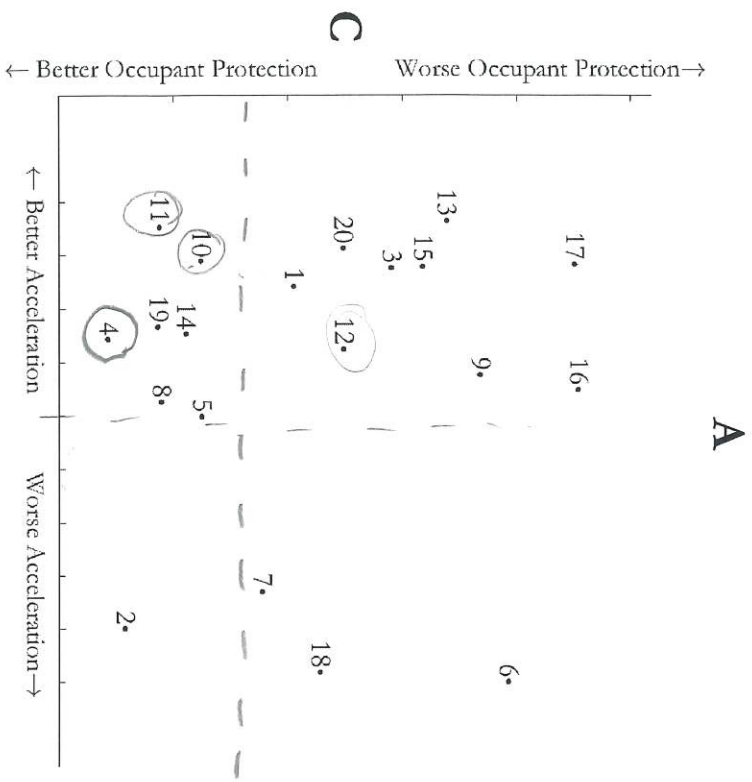
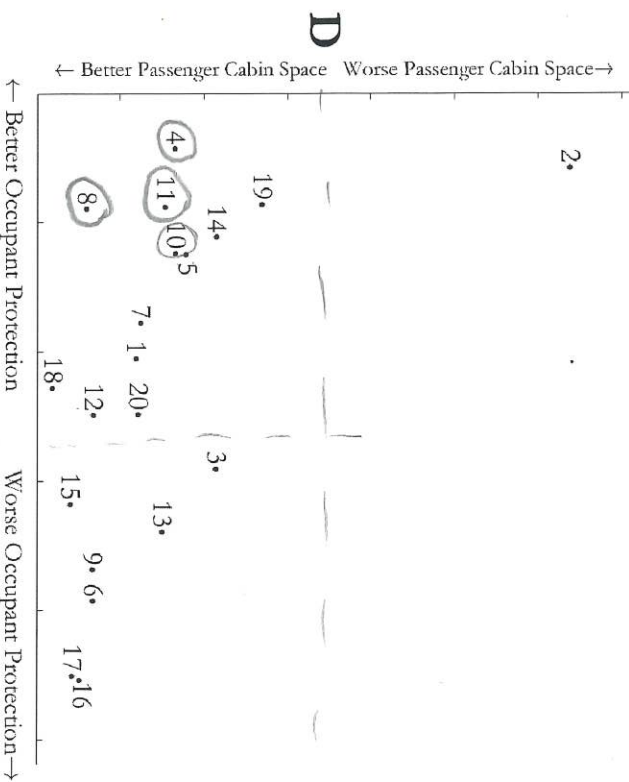
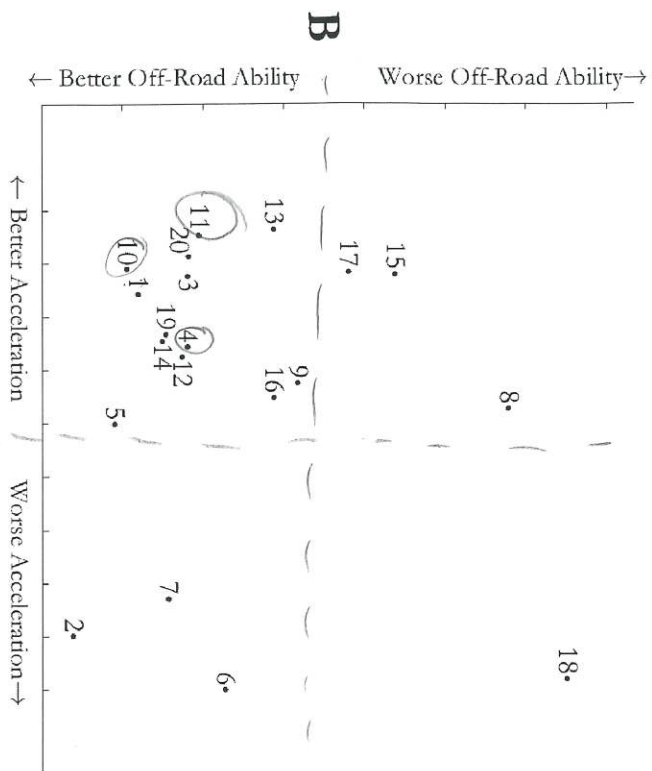
B

OFF:

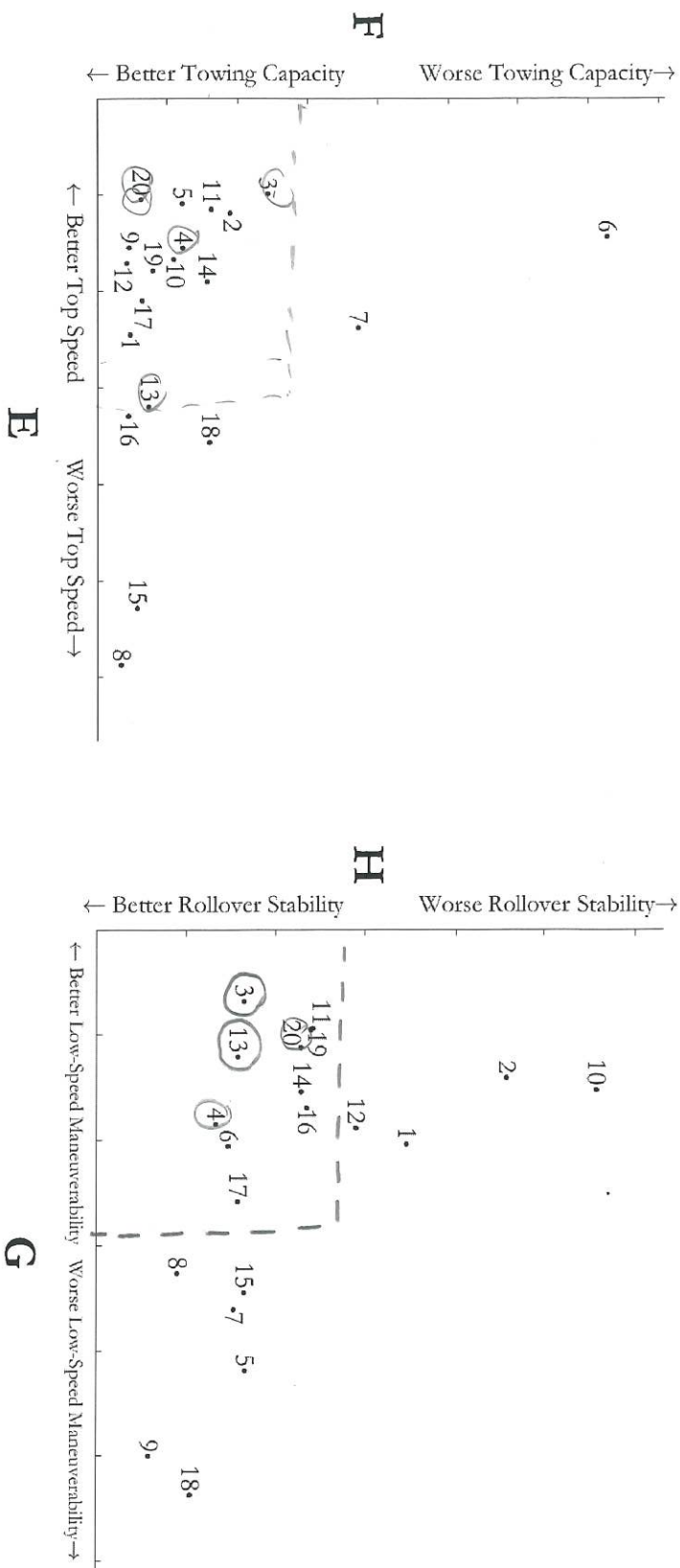
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Write your Group ID: 6000

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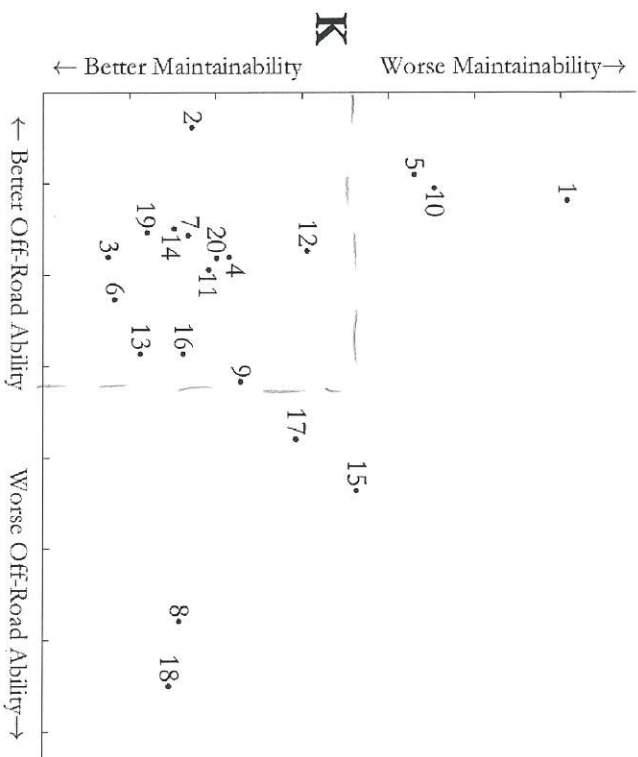
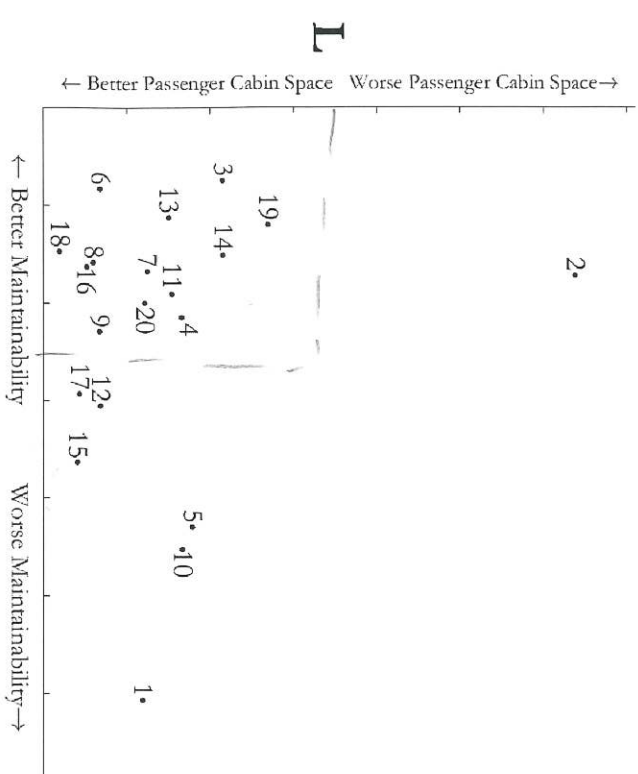
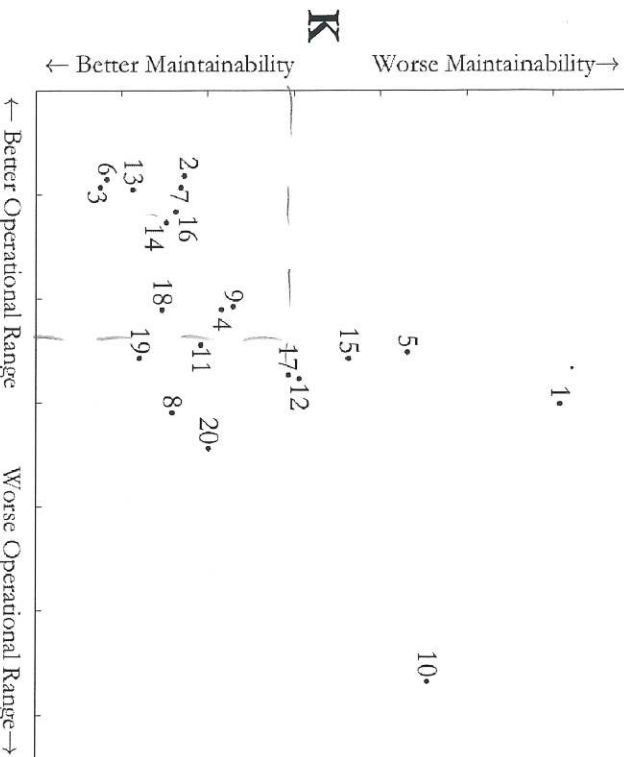
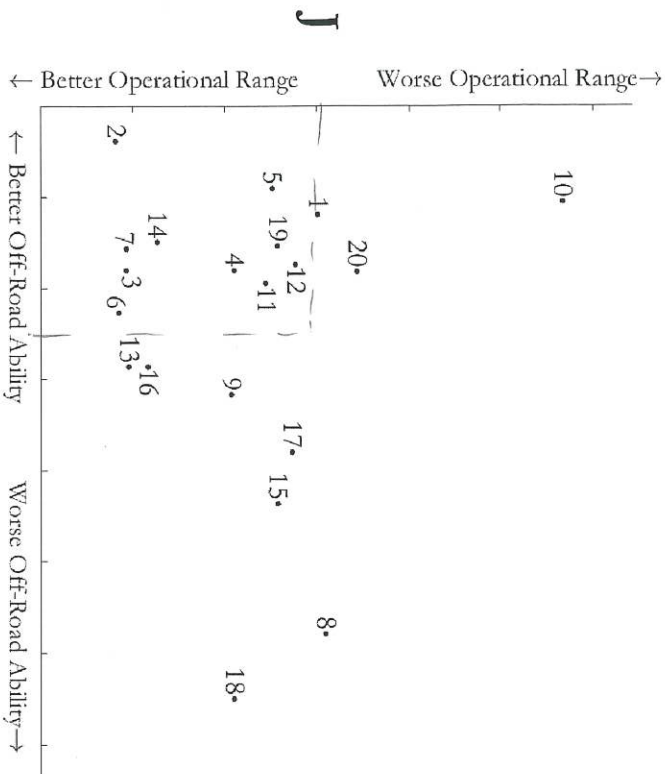


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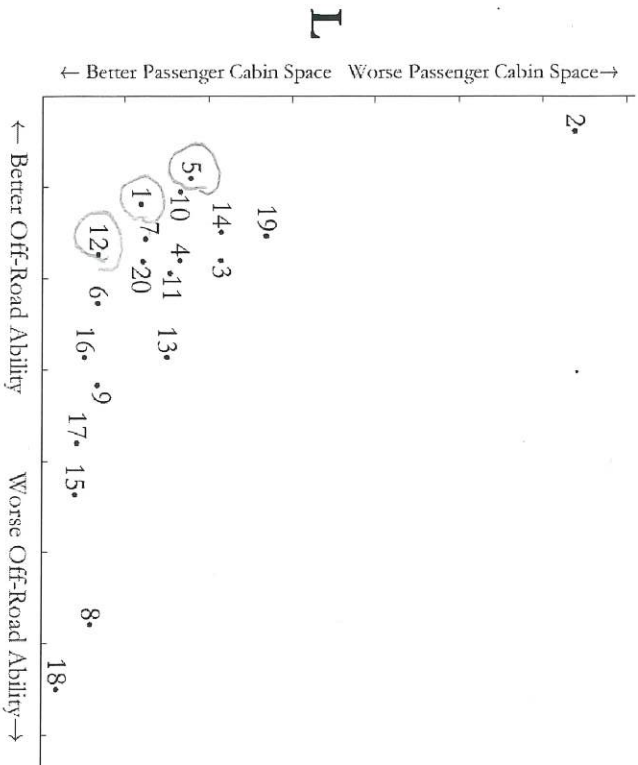
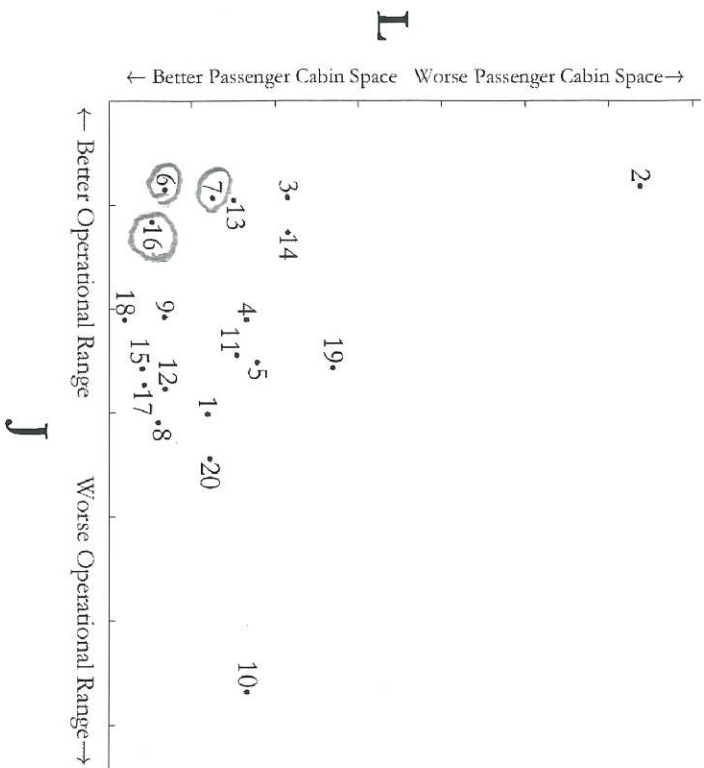


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Enter the number ID of the selected design: 6

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